Lab 1

```c
#include <stdio.h>
int main() {
   float Y_1, Y, I, T, C[5];
   float G[5] = {20, 25, 30, 35, 40};
   int i;
   printf("compiled by Aswin phuyal\n");
   printf("Enter initial value of lagged variable Y_1: ");
   scanf("%f", &Y_1);
   printf("\nThe growth in consumption is given in the following table:\n\n");
   printf("Year \t Consumption\n");

   for (i = 0; i < 5; i++) {
      I = 2 + 0.1 * Y_1;
      Y = 45.45 + 2.27 * (I + G[i]);
      T = 0.2 * Y;
      C[i] = 20 + 0.7 * (Y - T);
      printf("%d \t %.2f\n", i + 1, C[i]);
      Y_1 = Y;
   }
   return 0;
}
```

```
PS C:\Users\Aswin phuyal\Desktop\projects\webpractical\5th practical> cd "c:\Users\Asw
al\5th practical\simulation\" ; if ($?) { g++ 1.cpp -o 1 } ; if ($?) { .\1 }
compiled by Aswin phuyal
Enter initial value of lagged variable Y_1: 50

The growth in consumption is given in the following table:

Year     Consumption
1        79.77
2        93.34
3        102.78
4        111.28
5        119.56
PS C:\Users\Aswin phuyal\Desktop\projects\webpractical\5th practical\simulation>
```

Lab 2

```
#include <stdio.h>
int main() {
float l, m, e;
float p, et;
printf("compiled by Aswin phuyal\n");
printf("Enter Inter arrival time of customers per hour: ");
scanf("%f", &l);
printf("Enter The average time spent by customers per hour (service rate): ");
scanf("%f", &m);
p = 1 - (l / m);
e = l / (m - l);
et = (1 / (m - l)) * 60;
printf("\nThe probability that a customer won't have to wait atcounter: %f\n",
p);
printf("\nExpected number of customers in the bank: %f\n", e);
printf("\nExpected time to be spent in bank: %f minutes\n", et);
return 0;
}
```

```
PS C:\Users\Aswin phuyal\Desktop\projects\webpractical\5th practical> cd "c:\Users\Aswin phuyal\
al\5th practical\simulation\" ; if ($?) { g++ 2.cpp -o 2 } ; if ($?) { .\2 }
compiled by Aswin phuyal
Enter Inter arrival time of customers per hour: 12
Enter The average time spent by customers per hour (service rate): 18

The probability that a customer won't have to wait atcounter: 0.333333

Expected number of customers in the bank: 2.000000

Expected time to be spent in bank: 10.000000 minutes
PS C:\Users\Aswin phuyal\Desktop\projects\webpractical\5th practical\simulation>
```

Lab 4

```c
#include <stdio.h>
#include <math.h>
int main() {
float l, pr, a;
int x, i;
printf("compiled by Aswin phuyal\n");
printf("Enter arrival rate (lambda): ");
scanf("%f", &l);
printf("\nPoisson Distribution Probabilities:\n");
for (x = 0; x < 16; x++) {
a = 1;
for (i = 1; i <= x; i++) {
a *= i;}
pr = (exp(-l) * pow(l, x)) / a;
printf("P(X=%d) = %.6f\n", x, pr);
}
return 0;
}
```

```
PS C:\Users\Aswin phuyal\Desktop\projects\webpractical\5th practical> cd "c:\Users\Asw
al\5th practical\simulation\" ; if ($?) { g++ 4.cpp -o 4 } ; if ($?) { .\4 }
compiled by Aswin phuyal
Enter arrival rate (lambda): 3

Poisson Distribution Probabilities:
P(X=0) = 0.049787
P(X=1) = 0.149361
P(X=2) = 0.224042
P(X=3) = 0.224042
P(X=4) = 0.168031
P(X=5) = 0.100819
P(X=6) = 0.050409
P(X=7) = 0.021604
P(X=8) = 0.008102
P(X=9) = 0.002701
P(X=10) = 0.000810
P(X=11) = 0.000221
P(X=12) = 0.000055
P(X=13) = 0.000013
P(X=14) = 0.000003
P(X=15) = 0.000001
PS C:\Users\Aswin phuyal\Desktop\projects\webpractical\5th practical\simulation>
```

Lab 5

```
#include <stdio.h>
int main() {
float transMat[2][2] = {{0.9, 0.1}, {0.5, 0.5}};
float vect[1][2] = {{1.0, 0.0}};
float result[1][2] = {0.0, 0.0};
int i, j, k;
for (i = 0; i < 1; i++) {
for (j = 0; j < 2; j++) {
for (k = 0; k < 2; k++) {
result[i][j] += vect[i][k] * transMat[k][j];
}}}
printf("compiled by Aswin phuyal\n");
printf("\nWeather of next day using Markov chain:\n");
for (i = 0; i < 1; i++) {
for (j = 0; j < 2; j++) {
printf("%f\t", result[i][j]);
}
printf("\n");
}
return 0;
}
```

```
PS C:\Users\Aswin phuyal\Desktop\projects\webpractical\5th practical> cd "c:\Users\Aswin phuyal\
al\5th practical\simulation\" ; if ($?) { g++ tempCodeRunnerFile.cpp -o tempCodeRunnerFile } ; i
le }
compiled by Aswin phuyal

Weather of next day using Markov chain:
0.900000        0.100000
PS C:\Users\Aswin phuyal\Desktop\projects\webpractical\5th practical\simulation>
```

Lab 6

```
#include <stdio.h>
#define SIZE 100
int main() {
int x[SIZE];
int m = 100, a = 5, c = 13;
int i;
x[0] = 11;
for (i = 0; i < SIZE - 1; i++) {
x[i + 1] = (a * x[i] + c) % m;
}
printf("compiled by Aswin phuyal\n");
printf("Generation of random numbers using linear congruential method:\n");
for (i = 0; i < SIZE; i++) {
printf("%d\t", x[i]);
}
return 0;
}
```

```
PS C:\Users\Aswin phuyal\Desktop\projects\webpractical\5th practical\simulation> cd "c:\Users\Aswin phuyal\Desktop\projects\webpracti
cal\5th practical\simulation\" ; if ($?) { g++ 6.cpp -o 6 } ; if ($?) { .\6 }
compiled by Aswin phuyal
Generation of random numbers using linear congruential method:
11      68      53      78      3       28      53      78      3       28      53      78      3       28      53      78      3   2
8       53      78      3       28      53      78      3       28      53      78      3       28      53      78      3       28  5
3       78      3       28      53      78      3       28      53      78      3       28      53      78      3       28      53  7
8       3       28      53      78      3       28      53      78      3       28      53      78      3       28      53      78  3
28      53      78      3       28      53      78      3       28      53      78      3       28      53      78      3       28  5
3       78      3       28      53      78      3       28      53      78      3       28      53      78
PS C:\Users\Aswin phuyal\Desktop\projects\webpractical\5th practical\simulation>
```

Lab 7

```
#include <stdio.h>
#define SIZE 100
int main() {
long x[SIZE];
long m = 1000 , a = 15, c = 7;
int i;
x[0] = 13;
for (i = 0; i < SIZE - 1; i++) {
x[i + 1] = (a * x[i] + c) % m;
}
printf("compiled by Aswin phuyal\n");
printf("Generated 100 random numbers:\n\n");
for (i = 0; i < SIZE; i++) {
printf("%ld ", x[i]);
}
return 0;
}
```

```
PS C:\Users\Aswin phuyal\Desktop\projects\webpractical\5th practical> cd "c:\Users\Aswin phuyal\Desktop\projects\webpractical\5th
ctical\simulation\" ; if ($?) { g++ tempCodeRunnerFile.cpp -o tempCodeRunnerFile } ; if ($?) { .\tempCodeRunnerFile }
compiled by Aswin phuyal
Generated 100 random numbers:

13 202 37 562 437 562 437 562 437 562 437 562 437 562 437 562 437 562 437 562 437 562 437 562 437 562 437 562 437 562 437 562 437
 437 562 437 562 437 562 437 562 437 562 437 562 437 562 437 562 437 562 437 562 437 562 437 562 437 562 437 562 437 562 437 562 4
562 437 562 437 562 437 562 437 562 437 562 437 562 437 562 437 562 437 562 437 562 437 562 437 562 437 562 437 562 437 562 437 5
PS C:\Users\Aswin phuyal\Desktop\projects\webpractical\5th practical\simulation>
```

Lab 3

```c
#include <stdio.h>
int main() {
float arrival_rate = 1.0;
float service_time_sec = 20.0;
float service_rate = 60.0 / service_time_sec;
float travel_to_seat_min = 1.5;
float arrival_before_kickoff = 2.0;
float time_in_system_min = 1.0 / (service_rate - arrival_rate);
float total_time_min = time_in_system_min + travel_to_seat_min;
printf("compiled by Aswin phuyal\n");
printf("Total time to reach seat: %.2f minutes\n", total_time_min);
if (total_time_min <= arrival_before_kickoff) {
printf("Result: Yes, the fan will be seated before kick-off.\n");
} else {
printf("Result: No, the fan will miss the start of the game.\n");
}
return 0;
}
```

```
PS C:\Users\Aswin phuyal\Desktop\projects\webpractical\5th practical\simulation> cd "c:
cal\5th practical\simulation\" ; if ($?) { g++ 3.cpp -o 3 } ; if ($?) { .\3 }
compiled by Aswin phuyal
Total time to reach seat: 2.50 minutes
Result: No, the fan will miss the start of the game.
PS C:\Users\Aswin phuyal\Desktop\projects\webpractical\5th practical\simulation>
```

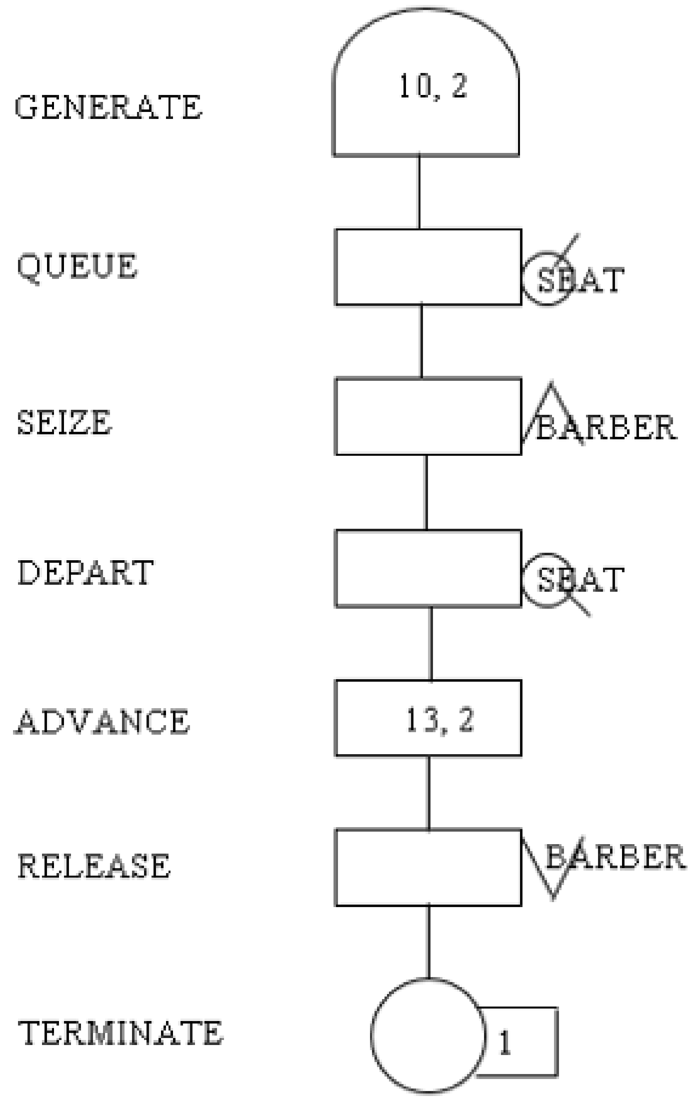
GPSS Questions:

Create a GPSS model and program to simulate a barber shop for a day (9 a.m. to 4 p.m.), where a customer enters the shop every 10 ± 2 minutes and a barber takes 13 ± 2 minutes for a haircut.

```
GENERATE 10,2
QUEUE SEAT
SEIZE BARBER
DEPART SEAT
ADVANCE 13,2
RELEASE BARBER
TERMINATE 1

GENERATE 420
TERMINATE 1

START 1
```



GPSS World Simulation Report - Untitled Model 1.1.1

Friday, June 19, 2026 12:57:01

START TIME	END TIME	BLOCKS	FACILITIES	STORAGES
0.000	25.893	9	1	0

NAME	VALUE
BARBER	10001.000
SEAT	10000.000

LABEL	LOC	BLOCK TYPE	ENTRY COUNT	CURRENT COUNT	RETRY
	1	GENERATE	2	0	0
	2	QUEUE	2	0	0
	3	SEIZE	2	1	0
	4	DEPART	1	0	0
	5	ADVANCE	1	0	0
	6	RELEASE	1	0	0
	7	TERMINATE	1	0	0
	8	GENERATE	0	0	0
	9	TERMINATE	0	0	0

FACILITY	ENTRIES	UTIL.	AVE. TIME	AVAIL.	OWNER	PEND	INTER	RETRY	DELAY
BARBER	2	0.561	7.262	1	3	0	0	0	0

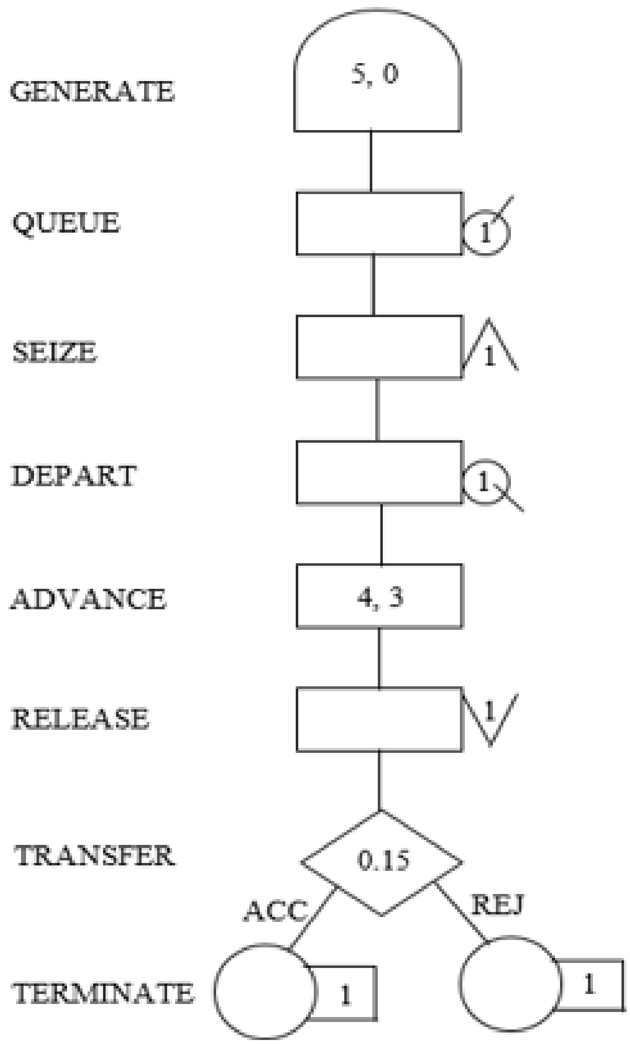
QUEUE	MAX	CONT.	ENTRY	ENTRY(0)	AVE.CONT.	AVE.TIME	AVE.(-0)	RETRY
SEAT	1	1	2	1	0.132	1.707	3.413	0

CEC XN	PRI	M1	ASSEM	CURRENT	NEXT	PARAMETER	VALUE
3	0	22.480	3	3	4		

FEC XN	PRI	BDT	ASSEM	CURRENT	NEXT	PARAMETER	VALUE
4	0	31.522	4	0	1		
2	0	420.000	2	0	8		

A machine tool in a manufacturing shop is turning out parts at the rate of every 5 minutes. When they are finished, the parts are sent to an inspector, who takes 4±3 minutes to examine each one and rejects 15% of the parts. Draw and explain a block diagram for it and write a GPSS program to simulate using the concept of FACILITY.

```
GENERATE 5,0
QUEUE INSQ
SEIZE INSPECTOR
DEPART INSQ
ADVANCE 4,3
RELEASE INSPECTOR
TRANSFER 0.15,REJ,ACC
ACC TERMINATE 1
REJ TERMINATE 1
```



GPSS World Simulation Report - Inspector.5.1

Thursday, June 18, 2026 16:18:51

START TIME	END TIME	BLOCKS	FACILITIES	STORAGES
0.000	11.286	11	1	0

NAME	VALUE
ACC	8.000
INSPECTOR	10001.000
INSQ	10000.000
REJ	9.000

LABEL	LOC	BLOCK TYPE	ENTRY COUNT	CURRENT COUNT	RETRY
	1	GENERATE	2	0	0
	2	QUEUE	2	0	0
	3	SEIZE	2	1	0
	4	DEPART	1	0	0
	5	ADVANCE	1	0	0
	6	RELEASE	1	0	0
	7	TRANSFER	1	0	0
ACC	8	TERMINATE	0	0	0
REJ	9	TERMINATE	1	0	0
	10	GENERATE	0	0	0
	11	TERMINATE	0	0	0

FACILITY	ENTRIES	UTIL.	AVE. TIME	AVAIL.	OWNER	PEND	INTER	RETRY	DELAY
INSPECTOR	2	0.557	3.143	1	3	0	0	0	0

QUEUE	MAX	CONT.	ENTRY	ENTRY(0)	AVE.CONT.	AVE.TIME	AVE.(-0)	RETRY
INSQ	1	1	2	1	0.114	0.643	1.286	0

CEC XN	PRI	M1	ASSEM	CURRENT	NEXT	PARAMETER	VALUE
3	0	10.000	3	3	4		

FEC XN	PRI	BDT	ASSEM	CURRENT	NEXT	PARAMETER	VALUE
4	0	15.000	4	0	1		
2	0	480.000	2	0	10		

A GENERATE block is used to represent the output of the machine by creating one transaction every five units of time. A QUEUE block places the transaction in the queue and SEIZE block allows a transaction to engage a facility if it is available. If more than one inspector is available, the transaction leaves the queue which is denoted by DEPART block and enters into ADVANCE block. An ADVANCE block with a mean of 4 and modifier of 3 is used to represent inspection. The time spent on inspection will therefore be any one of the values 1, 2, 3, 4, 5, 6 or 7, with equal probability given to each value. Upon completion of the inspection, RELEASE block allows a transaction to disengage the facility and transaction go to a TRANSFER block with a selection factor of 0.15, so that 85% of the parts go to the next location called ACC, to represent accepted parts and 15% go to another location called REJ to represent rejects. Both locations reached from the TRANSFER block are TERMINATE blocks.

```
SIMULATE

GENERATE 5,0,,,100
QUEUE INSQ
SEIZE INSPECTOR
DEPART INSQ
ADVANCE 4,3
RELEASE INSPECTOR
TRANSFER 0.15,REJ,ACC

ACC TERMINATE 1

REJ TERMINATE 1

START 100
```

File Edit Search View Command Window Help

GPSS World Simulation Report - Untitled Model 2.1.1

Friday, June 19, 2026 12:59:28

START TIME	END TIME	BLOCKS	FACILITIES	STORAGES
0.000	505.643	9	1	0

NAME	VALUE
ACC	8.000
INSPECTOR	10001.000
INSQ	10000.000
REJ	9.000

LABEL	LOC	BLOCK TYPE	ENTRY COUNT	CURRENT COUNT	RETRY
	1	GENERATE	101	0	0
	2	QUEUE	101	0	0
	3	SEIZE	101	1	0
	4	DEPART	100	0	0
	5	ADVANCE	100	0	0
	6	RELEASE	100	0	0
	7	TRANSFER	100	0	0
ACC	8	TERMINATE	17	0	0
REJ	9	TERMINATE	83	0	0

FACILITY	ENTRIES	UTIL.	AVE. TIME	AVAIL.	OWNER	PEND	INTER	RETRY	DELAY
INSPECTOR	101	0.826	4.133	1	101	0	0	0	0

QUEUE	MAX	CONT.	ENTRY	ENTRY(0)	AVE.CONT.	AVE.TIME	AVE.(-0)	RETRY
INSQ	2	1	101	42	0.206	1.029	1.762	0

CEC XN	PRI	M1	ASSEM	CURRENT	NEXT	PARAMETER	VALUE
101	100	505.000	101	3	4		

FEC XN	PRI	BDT	ASSEM	CURRENT	NEXT	PARAMETER	VALUE
102	100	510.000	102	0	1		

Parts are being made at the rate of one every 10 minutes. They are of two types; A and B. Mixed randomly with 10% being type B. Separate inspectors inspect each type. Inspection time for A is 6±2 minutes and for B is 10±2 minutes. Both inspectors reject 10% of parts. Simulate for 100 parts.

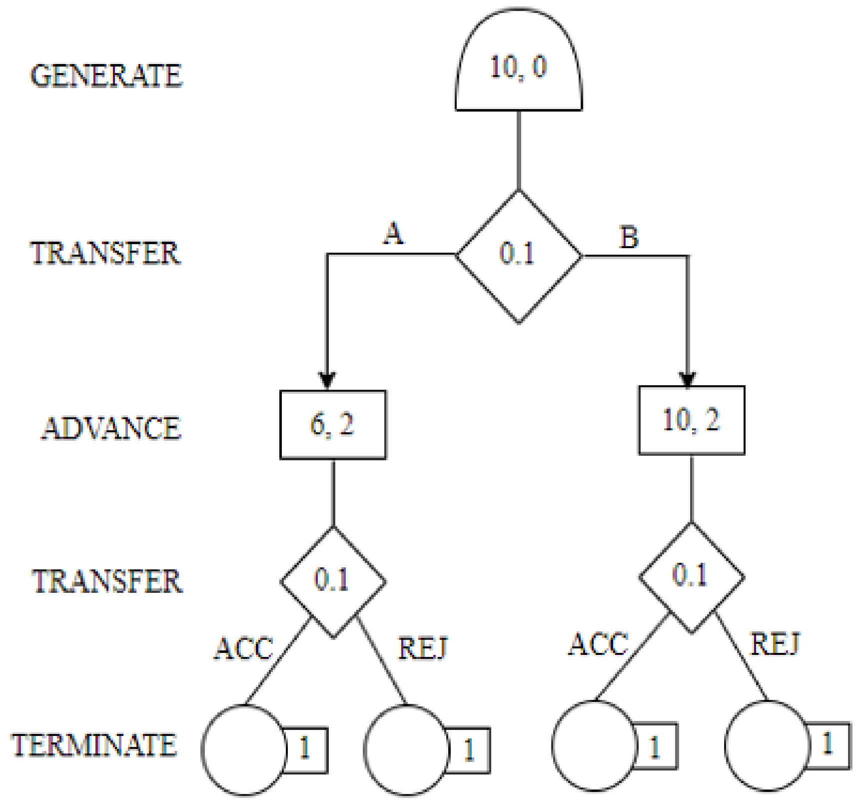
```
GENERATE 10,,,100
TRANSFER .10,TYPEB,TYPEA

TYPEA QUEUE AQUEUE
     SEIZE INSPA
     DEPART AQUEUE
     ADVANCE 6,2
     RELEASE INSPA
     TRANSFER .10,REJA,ACCA

ACCA  TERMINATE 1
REJA  TERMINATE 1

TYPEB QUEUE BQUEUE
     SEIZE INSPB
     DEPART BQUEUE
     ADVANCE 10,2
     RELEASE INSPB
     TRANSFER .10,REJB,ACCB

ACCB  TERMINATE 1
REJB  TERMINATE 1
START 100
```



File Edit Search View Command Window Help

GPSS World Simulation Report - Untitled Model 3.1.1

Friday, June 19, 2026 13:01:07

START TIME	END TIME	BLOCKS	FACILITIES	STORAGES
0.000	1008.573	18	2	0

NAME	VALUE
ACCA	9.000
ACCB	17.000
AQUEUE	10002.000
BQUEUE	10000.000
INSPA	10003.000
INSPB	10001.000
REJA	10.000
REJB	18.000
TYPEA	3.000
TYPEB	11.000

LABEL	LOC	BLOCK TYPE	ENTRY COUNT	CURRENT COUNT	RETRY
	1	GENERATE	100	0	0
	2	TRANSFER	100	0	0
TYPEA	3	QUEUE	11	0	0
	4	SEIZE	11	0	0
	5	DEPART	11	0	0
	6	ADVANCE	11	0	0
	7	RELEASE	11	0	0
	8	TRANSFER	11	0	0
ACCA	9	TERMINATE	2	0	0
REJA	10	TERMINATE	9	0	0
TYPEB	11	QUEUE	89	0	0
	12	SEIZE	89	0	0
	13	DEPART	89	0	0
	14	ADVANCE	89	0	0
	15	RELEASE	89	0	0
	16	TRANSFER	89	0	0
ACCB	17	TERMINATE	11	0	0
REJB	18	TERMINATE	78	0	0

FACILITY	ENTRIES	UTIL.	AVE. TIME	AVAIL.	OWNER	PEND	INTER	RETRY	DELAY
INSPB	89	0.871	9.871	1	0	0	0	0	0
INSPA	11	0.075	6.849	1	0	0	0	0	0

QUEUE	MAX	CONT.	ENTRY	ENTRY(0)	AVE.CONT.	AVE.TIME	AVE.(-0)	RETRY
BQUEUE	1	0	89	37	0.115	1.300	2.225	0
AQUEUE	1	0	11	11	0.000	0.000	0.000	0